believe the welfare state is highly desirable. But you cannot have an unnatural welfare state, unless you also have unnatural birth control; otherwise, the end result will be misery even greater than that which obtains in nature. The welfare state is perhaps the greatest altruistic system the animal kingdom has ever known. But any altruistic system is inherently unstable, because it is open to abuse by selfish individuals, ready to exploit it. Individual humans who have more children than they are capable of rearing are probably too ignorant in most cases to be accused of conscious malevolent exploitation. Powerful institutions and leaders who deliberately encourage them to do so seem to me less free from suspicion.

Returning to wild animals, the Lack clutch size argument can be generalized to all the other examples Wynne-Edwards uses: territorial behaviour, dominance hierarchies, and so on. Take, for instance, the red grouse that he and his colleagues have worked on. These birds eat heather, and they parcel out the moors in territories containing apparently more food than the territory owners actually need. Early in the season they fight over territories, but after a while the losers seem to accept that they have failed, and do not fight any more. They become outcasts who never get territories, and by the end of the season they have mostly starved to death. Only territory owners breed. That non-territory owners are physically capable of breeding is shown by the fact that if a territory owner is shot his place is promptly filled by one of the former outcasts, who then breeds. Wynne-Edwards's interpretation of this extreme territorial behaviour is, as we have seen, that the outcasts 'accept' that they have failed to gain a ticket or licence to breed; they do not try to breed.

On the face of it, this seems an awkward example for the selfish gene theory to explain. Why don't the outcasts try, try, and try again to oust a territory holder, until they drop from exhaustion? They would seem to have nothing to lose. But wait, perhaps they do have something to lose. We have already seen that if a territory-holder should happen to die, an outcast has a chance of taking his place, and therefore of breeding. If the odds of an outcast's succeeding to a territory in this way are greater than the odds of his gaining one by fighting, then it may pay him, as a selfish individual, to wait in the hope that somebody will die, rather than squander what little energy he has in futile fighting. For Wynne-Edwards, the role of the outcasts in the welfare of the group is to wait in